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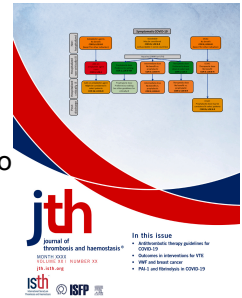
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Dysregulation of HIF-1 α -regulated and thrombo-inflammatory genes may contribute to thrombotic risk in patients with *BCR::ABL1*-negative myeloproliferative neoplasms

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Abstract

Background Hypoxia-inducible factor-1 α (HIF-1 α) regulates multiple pathways involved in metabolism, inflammation, and coagulation and may contribute to thrombotic complications in patients with *BCR::ABL1*-negative myeloproliferative neoplasms (MPN). This study investigated the expression of selected HIF-1 α -regulated and thrombo-inflammatory genes (*SLC2A1*, *F3*, *SELP*, *VEGFA*, *SERPINE1*, *PROS1*, *NLRP3* and *THBS1*) and their association with thrombotic events, *JAK2* mutation status, and cytoreductive therapy.

Methods Data of 120 MPN patients (40 with polycythemia vera – PV, 40 with essential thrombocythemia – ET, and 40 with primary myelofibrosis – PMF) and their blood samples were analyzed. In each MPN subtype, number 50% of patients had a history of thrombosis. RNA expression levels were measured using qRT-PCR.

Results RNA expression patterns differed across MPN subtypes. Higher *F3* expression in ET and increased *SLC2A1* and *SELP* expression in PV and PMF were observed in patients with thrombosis. In multivariable logistic regression, *SLC2A1* (OR 2.05, 95% CI 1.20 - 3.49, p = 0.008)

and *SELP* expression (OR 1.79, 95% CI 1.06 - 3.02, $p = 0.03$) remained independently associated with thrombotic events, whereas *NLRP3* expression was inversely associated (OR 0.58, 95% CI 0.37 - 0.90, $p = 0.02$). Most analyzed genes were more highly expressed in *JAK2*-mutated than in *JAK2*-negative patients. Cytoreductive therapy showed no significant impact on RNA expression.

Conclusion Subtype-specific RNA expression patterns may reflect differences in thrombotic incidence across MPN entities. Higher RNA expression in *JAK2*-mutated patients supports a link between *JAK2* signaling and HIF-related pathways. Further studies are warranted to clarify their role and therapeutic potential.

Keywords: myeloproliferative neoplasms, HIF-1 α -regulated genes, venous thrombosis, arterial thromboembolism

Introduction

According to the 2022 World Health Organization (WHO) classification for hematopoietic malignancies [1], major categories of *BCR::ABL1*-negative myeloproliferative neoplasms (MPN) include polycythemia vera (PV), essential thrombocythemia (ET), and primary myelofibrosis (PMF). MPN are associated with an increased thrombotic risk, with a reported incidence ranging from 20% [2] to 33.6% [3]. Thrombotic complications significantly contribute to morbidity and mortality in this patient group [4 - 6]. Moreover, patients with venous thrombosis in unusual sites represent a distinct cohort with unique molecular and clinical characteristics, with a 5-year mortality of about 40% [7 - 9]. The role of age > 60 years, prior thrombosis, cardiovascular risk factors, *JAK2* mutation (including high *JAK2*-V617F allele frequency) and non-driver mutations (e.g., *TET2* and *DNMT3A*), thromboinflammation, blood cell counts, and their dysfunctions have been described as associated with an increased thrombotic risk in MPN patients [3, 10 - 23]. Despite modern cytoreductive therapy, the frequency of thrombotic complications continues to increase over time after an MPN diagnosis [24-26].

Hypoxia-inducible factors (HIF) are transcriptional regulators of an extensive spectrum of target genes that play a role in angiogenesis, metabolism, and thromboinflammation [27 - 28]. According to the study by Gangaraju et al. [29], granulocytes and platelets from patients with PV and ET had higher levels of transcripts of several thrombotic and inflammatory HIF-regulated genes (*F3*, *SELP*, *VEGFA*, *SLC2A1*, and *IL1RAP*) than controls. Moreover, the expression levels

of these genes were significantly increased in patients with a history of thrombosis, compared to patients without thrombosis. Another study [30] has analyzed the expression of HIF-1-regulated genes (*SLC2A1*, *PDK1*, *VEGFA*, *NFE-2*, and *RUNX2*) in granulocytes and platelets from MPN patients and identified upregulation of *SLC2A1* and *PDK1* transcripts, as well as increased glucose uptake in granulocytes, suggesting the presence of a Warburg-like metabolic shift. A recent investigation by Baumeister et al. [31, 32] has found that HIF-1 signaling significantly contributes to *JAK2-V617F*-positive MPN pathogenesis, and the inhibition of HIF-1 α precisely impaired the proliferation and survival of *JAK2-V617F*-mutated cells. The study by Ghosh et al. [33] demonstrates in a *JAK2-V617F*-driven mouse model that increased HIF- α expression is associated with a markedly enhanced thrombotic phenotype, and pharmacological inhibition of HIF-1 α or HIF-2 α significantly reduced thrombus size. Furthermore, treatment with the JAK inhibitors ruxolitinib or fedratinib, in the presence of TNF- α , significantly reduced *F3* mRNA upregulation [34]. Importantly, Kealy et al. [35] demonstrated that *JAK2-V617F* activates a distinct HIF-1 transcriptional program through a PIM1-dependent mechanism that differs from canonical hypoxia signaling and is associated with disease severity, progression, and survival in MPN. These findings underscore the role of HIF-1 α as a potential contributor to disease progression and suggest it may represent a promising therapeutic target in MPN.

A detailed understanding of pathophysiology of thrombotic complications is essential for risk assessment, developing targeted therapies and reduction of mortality in MPN patients. Hence, this study aims to investigate the dysregulation of HIF1 α -regulated and thrombo-inflammatory genes *SLC2A1*, *F3*, *SELP*, *VEGFA*, *SERPINE1*, *PROS1*, *NLRP3* and *THBS1* in patients with prior thrombosis compared to those without. Furthermore, it attempts to assess the correlation between the expression levels of these genes and both the *JAK2* mutation status and *JAK2-V617F* allele frequency (VAF). Additionally, the potential influence of cytoreductive therapy on the expression levels of these genes has been evaluated.

Methods

Patients' cohort

In total, 120 MPN patients (40 with PV, 40 with ET, and 40 with PMF) treated between 2005 and 2024 at the Jena University Hospital were analyzed. Among the 40 patients with PMF, 20 had prefibrotic myelofibrosis. Patients with secondary myelofibrosis were not included. Blood samples were collected at the time of the first presentation at the hematology department of Jena University

Hospital and stored in the biobank after processing. At the time of sampling, 40 patients were already receiving cytoreductive therapy, whereas the remaining 80 patients were newly diagnosed and treatment-naïve, indicating a heterogeneous treatment status at sampling. A history of at least one thrombotic event was observed in 50% of patients in each MPN subtype (corresponding to 60 out of 120 patients in total): patients have been deliberately selected to include equal representation of those with and without thrombotic event, consistent with a case-control-type design to facilitate comparative molecular analyses. A thrombotic event was diagnosed at the time of MPN diagnosis, or it occurred prior to MPN diagnosis, with a median of 21 months (range, 5 - 48 months). Five patients had diabetes mellitus (three treatment-naïve, two on cytoreductive therapy), which may influence *SLC2A1* expression and was therefore considered in the statistical analysis. The study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Ethics Committee of the University Jena.

Gene panel for expression analysis

A targeted panel of genes involved in thrombo-inflammatory and HIF-1 α -related pathways was selected for the current study, including tissue factor (*F3*), P-selectin (*SELP*), serpin peptidase inhibitor clade E member 1 (*SERPINE1*), which encodes plasminogen activator inhibitor-1 (PAI-1), thrombospondin 1 (*THBS1*), protein S (*PROS1*), two HIF-regulated genes (vascular endothelial growth factor A (*VEGFA*) and solute carrier family 2 member 1 (*SLC2A1*), a facilitated glucose transporter, and NLR family pyrin domain-containing 3 (*NLRP3*), encoding a key component of the inflammasome that mediates IL-1 β - and IL-18-driven inflammation and contributes to thrombo-inflammatory processes. These genes were selected based on published data on hypoxia signaling and thrombotic mechanisms, as well as their potential biological relevance in *BCR::ABL1*-negative MPN [29, 31-39]. Although not complete, the panel was designed to cover a broad spectrum of functionally relevant pathways.

Leukocyte isolation and nucleic acid preparation

Leukocytes were isolated from peripheral blood samples by selective erythrocyte lysis. Approximately 10 mL of patient material was transferred to 50 mL tubes, diluted to 50 mL with erythrocyte lysis buffer, and mixed gently. The tubes were incubated on ice for 10 min with intermittent mixing, followed by centrifugation for 10 min at 400 $\times$ g and 4 °C. If the pellet

appeared white, processing continued, otherwise, the lysis step was repeated. The supernatant was discarded, and 25 mL of phosphate-EDTA buffer was added. The pellet was carefully resuspended and centrifuged again for 5 min at $400 \times g$ and 4°C . The supernatant was discarded, and the pellet was resuspended in approximately 1 mL of phosphate-EDTA buffer. Cell counts were determined using a Countess 3 Automated Cell Counter (Thermo Fisher, Waltham, USA). For further analysis, 1×10^7 cells were transferred into preservation tubes and centrifuged at $1,500 \times g$ for 3 min and processed for DNA or RNA isolation. DNA from peripheral blood was isolated using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's recommendations. For RNA isolation, 500 μL of TRIzolTM (Thermo Fisher, Waltham, MA, USA) reagent was added to each pellet, followed by homogenization using a syringe and needle, and the samples were then stored at -20°C until RNA extraction according to manufacturer's instructions. RNA concentration and purity was determined using a DeNovix DS-11 (DeNovix, Wilmington, Delaware, USA) spectrophotometer. Given the instability of RNA, cDNA synthesis was performed immediately after isolation using SuperScriptTM IV VILOTM Master Mix (Thermo Fisher Scientific, Waltham, MA, USA).

Mutation and RNA expression analysis

Mutation and expression analyses were performed for all 120 patients. The hotspot *JAK2-V617F* mutation was detected using the amplification-refractory mutation system (JAK2-ARMS-PCR, Mastercycler[®] X50, Eppendorf, Hamburg, Germany) [40]. The *JAK2-V617F* VAF was assessed in all patients by pyrosequencing as described previously [41]. *JAK2-V617F* mutation was considered positive for $\text{VAF} \geq 5.0\%$. To analyze *CALR* and *MPL* mutations, DNA fragment analysis and Sanger sequencing were performed on the 3500 Genetic Analyzer, respectively (ThermoFisher Scientific, Waltham, MA, USA) [42]. Expression of selected HIF-1 α -regulated and thrombo-inflammatory genes was analyzed using quantitative real-time polymerase chain reaction (qRT-PCR; PowerTrackTM SYBR Green Master Mix, ThermoFisher Scientific) on the QuantStudioTM 5 Real-Time PCR System (Thermo Fisher Scientific, Waltham, MA, USA). Relative gene expression was determined using a comparative quantification approach based on the Pfaffl method [43]. Primer sequences were selected based on established qRT-PCR assays, including publicly available primer sets [44], or optimized during initial validation experiments (final primer pairs were selected based on optimal amplification performance and specificity). All

assays were conducted under standardized conditions using the same platform and reagents to ensure comparability. *RPL13A* was used as the housekeeping gene due to its reported high expression stability under hypoxic conditions and relatively consistent expression across different leukocyte subpopulations [29, 45 - 47]. It was included for each target gene, sample, and run, and was used for normalization of expression levels. All reactions were performed in technical replicates, and repeated if the standard deviation (SD) was greater than 1.0 Ct. Primer sequences are provided in Supplementary Table 1.

Statistics

A statistical comparison was made between the equal-number groups of patients with thrombosis ($n = 60$) and those without ($n = 60$). Usual site venous thrombotic events (VTE) include deep vein thrombosis and pulmonary embolism; unusual site VTE include splanchnic vein thrombosis (SVT) and cerebral venous sinus thrombosis. Arterial thrombotic events (ATE) include ischemic stroke or transient ischemic attack, peripheral vascular disease, or acute coronary syndrome. Variables were reported as median with interquartile range (IQR). *JAK2*-mutated patients were stratified into VAF quartiles (Q1: 5.3-17.2%, Q2: 17.3-24.0%, Q3: 24.1-38.1%, Q4: 38.2-90.2%) based on the distribution of observed allele frequencies. The medians of the two groups were compared using the Wilcoxon-Mann-Whitney test. Spearman's rank correlation was used for correlation analyses due to non-normal data distribution. Categorical variables were reported as counts and proportions and compared using Chi-square test. All reported p-values are two-sided, and $p < 0.05$ was considered statistically significant. To account for multiple comparisons across the analyzed genes, p values were adjusted using the Benjamini-Hochberg false discovery rate (FDR) procedure. An adjusted $q < 0.05$ was considered statistically significant. Multivariable logistic regression analysis was performed to identify genes independently related to thrombotic events. A subgroup analysis was performed to evaluate potential differences between treatment-naïve patients and those receiving cytoreductive therapy. To evaluate the impact of cytoreductive therapy on RNA expression levels, multivariable linear regression including MPN subtype was applied. SPSS Statistics version 28.0 was used for statistical analyses, and GraphPad Prism version 10.2.3 was used for data visualization and figure generation.

Results

In total, 120 patients with MPN were analyzed: 40 with PV, 40 with ET, and 40 with PMF. The median age at diagnosis was 55.5 years for PV (range, 25-83), 48.0 years for ET (range, 18-81), and 63.0 years for PMF (range, 26-77). Overall, 59 patients (49.2%) were female. Mutation analysis was performed for all patients: 90/120 (75%) patients showed a *JAK2* mutation (89 patients with *JAK2-V617F* mutation and one patient with a mutation in exon 12 – p.I540_E543delinsML), 22/120 (18.4%) patients showed *CALR*, 4/120 (3.3%) patients showed an *MPL* mutation, 4 (3.3%) were triple negative. The median *JAK2-V617F* VAF was 27.8% (range, 5.0-90.2%). Among patients with VTE, 27 (71%) patients had a thrombosis at unusual sites. The baseline characteristics of the patients are summarized in Table 1.

Table 1. Characteristics of the study population, n = 120

Variable/ MPN subtype	Polycythemia vera, n = 40	Essential thrombocythemia, n = 40	Primary Myelofibrosis, n = 40
Driver mutation			
<i>JAK2-V617F</i> or exon 12 (n, %)	40 (100%)	27 (67.5%)	23 (57.5%)
<i>CALR</i> (n, %)	-	10 (25.0%)	12 (30.0%)
<i>MPL</i> (n, %)	-	2 (5.0%)	2 (5.0%)
triple-negative (n, %)	-	1 (2.5%)	3 (7.5%)
Thrombotic events	n = 20	n = 20	n = 20
arterial (n, %)	7 (35.0%)	8 (40.0%)	7 (35.0%)
venous (n, %)	13 (65.0%)	12 (60.0%)	13 (65.0%)
Therapy			
Anagrelide	1 (2.5%)	1 (2.5%)	-
Interferon alpha	2 (5.0%)	6 (15.0%)	1 (2.5%)
Hydroxycarbamide	7 (17.5%)	5 (12.5%)	8 (20.0%)
JAK inhibitor	3 (7.5%)	-	6 (15.0%)

Expression of HIF-1 α -regulated and thrombo-inflammatory genes in patients with thrombosis versus those without. The influence of cytoreductive therapy on RNA expression.

In PV, *SLC2A1* expression was higher in patients with thrombosis, but not after FDR correction ($p < 0.05$, $q = 0.18$). In ET, *F3* remained significantly associated with thrombosis after FDR correction ($p = 0.007$, $q = 0.05$). In PMF, *SELP* expression was higher in patients with thrombosis, but not after correction for multiple testing ($p < 0.05$, $q = 0.33$) (Figure 1). There were no statistically significant differences in the expression of the analyzed genes between patients with VTE and ATE. However, a significantly higher expression of *SLC2A1* ($p < 0.05$) was observed in patients with thrombosis at unusual sites. In a multivariable logistic regression model including selected genes and adjusting for therapy status and diabetes, higher expression of *SLC2A1* (OR 2.05, 95% CI 1.20 - 3.49, $p = 0.008$) and *SELP* (OR 1.79, 95% CI 1.06 - 3.02, $p = 0.03$) remained significant predictors of thrombosis. In contrast, *NLRP3* expression was inversely associated with thrombotic events (OR 0.58, 95% CI 0.37 - 0.90, $p = 0.02$). Linear regression analysis has been performed to evaluate the potential effect of therapy on RNA expression. Although nominal associations have been observed for *SERPINE1* ($p = 0.03$) and *PROS1* ($p = 0.02$), these did not remain significant after FDR correction ($q = 0.13$ for both)

Expression of HIF-1 α -regulated and thrombo-inflammatory genes according to JAK2 mutation status and JAK2-V617F VAF.

Thrombotic events have been found to occur more frequently in patients harboring *JAK2* mutations compared with patients without *JAK2* mutations ($p = 0.013$). Among *JAK2*-mutated patients stratified by *JAK2*-V617F VAF quartiles (Q1-Q4), thrombotic events occurred in 60.0% of patients in Q1, 84.2% in Q2, 35.0% in Q3, and 44.4% in Q4.

After FDR correction, all analyzed genes except *NLRP3* remained significantly associated with *JAK2* mutation status (*SERPINE1* $q = 0.02$, *PROS1* $q = 0.04$, *SELP* $q = 0.04$, *VEGFA* $q = 0.04$, *THBS1* $q = 0.05$, *SLC2A1* $q = 0.05$, *F3* $q = 0.05$) (Figure 2).

Discussion

Patients with MPN and thrombotic events exhibited selective upregulation of several HIF-1 α -regulated and thrombo-inflammatory genes, with distinct patterns across MPN subtypes, suggesting subtype-specific differences in the underlying pathophysiology.

In PV, only *SLC2A1* was significantly upregulated in patients with thrombosis. However, the association did not remain significant after FDR correction, although it may represent a biologically relevant signal. This exclusive overexpression may resemble a Chuvash polycythemia-like dysregulation, in which the *VHL-R200W* mutation inhibits degradation of HIF-1 α and HIF-2 α [48 - 50]. Consequently, it leads to accumulation of HIF activity and subsequently to upregulation of downstream target genes, including *SLC2A1*, *VEGFA*, *EPO* (encoding erythropoietin), and others, which are involved in oxygen transport, metabolism, erythropoiesis, and angiogenesis [51]. This result aligns with the Warburg effect, a unique metabolic phenomenon prevalent in cancer cells, with the preference for glycolysis to generate energy even under normoxic conditions [52, 53]. HIF-1 plays a pivotal role in this metabolic change, as demonstrated in several cancer types [54 - 56]. Additionally, a recent study [30] investigated the Warburg effect across MPN subtypes by analyzing the expression of HIF-1-regulated genes: an upregulation of *SLC2A1* transcripts and enhanced glucose uptake in leukocytes from MPN patients were found. In line with this hypothesis, multivariable logistic regression analysis identified higher *SLC2A1* expression as an independent predictor of thrombotic events. Furthermore, PV is associated with the highest thrombotic risk among the classical MPN subtypes, and *SLC2A1* upregulation was observed specifically in this subgroup.

The observed upregulation of *SLC2A1* in patients with thrombosis at unusual sites compared with patients with usual site VTE is of particular interest, as endothelial cells harboring the *JAK2-V617F* mutation in the splanchnic circulation may promote thrombotic processes via HIF-1 α accumulation [57, 58].

Mechanistically, HIF signaling regulates multiple prothrombotic pathways, including upregulation of tissue factor (*F3*), endothelial and platelet activation and aggregation (*SELP*, *THBS1*), inhibition of fibrinolysis and stabilization of clots (*SERPINE1*), and downregulation of natural anticoagulants such as protein S (*PROS1*). In the line with previous studies, in ET patients with thrombosis, higher expression of *F3* was observed, consistent with its central role in coagulation initiation [29, 39]. In the PMF cohort, *SELP* was upregulated in patients with thrombotic events, and multivariable logistic regression analysis further identified *SELP* as an independent predictor for thrombosis. *SELP* encodes a cell adhesion molecule that mediates leukocyte-endothelial-platelet interactions. Increased *SELP* expression has been associated with cardiovascular complications [60 - 62]. Furthermore, inhibition of P-selectin has demonstrated a

reduction in thrombosis within MPN models [57], and clinical evaluation is ongoing in patients with PMF (NCT04097821) [63]. Thus, the upregulation of *SELP* in thrombotic PMF cases highlights the role of platelet-leukocyte aggregates and endothelial activation in PMF-associated hypercoagulability.

Similar to the findings reported by Gangaraju et al. [29], no statistically robust evidence that cytoreductive therapy alters the expression of HIF-regulated genes was observed in the current study. In univariable analysis, *SERPINE1* and *PROS1*, which encode key regulators of the coagulation and fibrinolytic balance, showed lower expression levels in patients receiving therapy. These observations should be interpreted with caution given the limited statistical power and the heterogeneity of treatment regimens within the cohort. In contrast to previous report [39], no therapy-related differences in *F3* RNA expression were detected, which may reflect the low prevalence of JAK inhibitors in the present analysis. Further studies are required to investigate the impact of cytoreductive treatment on the expression of HIF-1 α -related genes in patients with MPN [32, 64].

Patients with *JAK2* mutations exhibited higher expression levels of all selected HIF-1 α -regulated and thrombo-inflammatory genes compared with *JAK2-V617F*-negative patients, with the exception of *NLRP3* (a non-significant trend). As has been well established, the presence of a *JAK2-V617F* mutation was associated with the occurrence of thrombotic events. Previous studies have shown that patients with *JAK2-V617F* mutation demonstrated the highest values of platelet surface tissue factor and P-selectin [65, 66]. *JAK2-V617F*-positive endothelial cells have been shown to promote a prothrombotic state due to increased P-selectin expression [57, 58]. In *BCR::ABL1*-negative MPN, the *JAK2-V617F* mutation leads to constitutive activation of JAK2 kinase, facilitating excessive myeloproliferation through enhanced PI3K-AKT-mTOR signaling and increased reactive oxygen radicals, and as a result, to elevated synthesis and stability of HIF-1 α even under normoxic conditions [27, 28, 67]. In the present cohort, no linear correlation between *JAK2-V617F* VAF and the occurrence of thrombotic events was observed. One possible explanation may be the high proportion of patients with thrombosis at unusual sites ($n = 27$, 71%) as well as the inclusion of patients with PMF, a subgroup in which data on the association between high *JAK2-V617F* VAF and thrombotic events are limited. Furthermore, one third of patients were already receiving cytoreductive therapy at the time of sampling, which may have reduced *JAK2-V617F* VAF [68, 69]. Importantly, *F3* expression showed a moderate inverse correlation with

JAK2-V617F VAF ($r = -0.45$). While *JAK2*-mutated MPN has previously been associated with increased tissue factor expression and activity in neutrophils (Reeves et al.), *F3* expression showed a moderate inverse correlation with *JAK2-V617F* VAF ($r = -0.45$) in the current study. This discrepancy may reflect differences in post-transcriptional regulation of tissue factor activity.

Recent analysis by Song et al. [70] showed that alternative splicing of *NFKB1*, particularly exon 4 or 5 skipping (AS-*NFKB1*), links NF- κ B signaling with inflammatory and HIF-regulated gene expression and correlates with increased hemoglobin levels in Andean Aymara highlanders, potentially conferring protection against excessive inflammation. These effects were independent of canonical *NFKB1* expression levels. While the current study focused on canonical inflammatory and hypoxia-associated transcriptional pathways, alternative splicing events of *NFKB1* were not assessed. Therefore, AS-*NFKB1*-mediated regulation may represent an additional mechanism contributing to the observed inflammatory and HIF-related signatures and warrants future investigations. Interestingly, *NLRP3* expression showed an inverse association with thrombotic risk in the current analysis. This finding should be interpreted with caution, as transcriptional levels of *NLRP3* may not directly reflect mechanistical impact to thromboinflammation. Furthermore, lower *NLRP3* protein expression is not uniformly “good” or “bad”: it tends to be beneficial in cancer and sepsis, where NLRP3-driven inflammation and proliferation are pathogenic, but in autoimmune diseases it appears to be linked to impaired host defense or a complex immune rebalancing.

Although the study did not resolve the question of whether the thrombosis in MPN is mostly or particularly caused by the involvement of HIF signaling, it demonstrated subtype-specific upregulation of several HIF-1 α -regulated and thrombo-inflammatory genes in patients with MPN and thrombosis, that warrant further investigation.

Conclusions

In conclusion, the current study provides evidence that the expression of selected HIF-1 α -regulated and thrombo-inflammatory genes differs between MPN patients with and without thrombotic events and varies according to disease subtype. Upregulation of *SLC2A1* and *SELP* was independently associated with thrombotic events, whereas *NLRP3* expression was inversely related to thrombosis. The expression of most analyzed genes was positively correlated with *JAK2* mutation status, supporting the link between *JAK2*-driven signaling and HIF pathway activation.

Cytoreductive therapy had no significant impact on gene expression. Overall, these findings suggest that HIF-1 α and its target genes may contribute to thrombotic risk in MPN.

Limitations and future perspectives

This study has several limitations. RNA expression was not assessed exactly at the time of the thrombotic event. Gene expression was analyzed in an unselected leukocyte population without adjustment for leukocyte subset composition, neutrophil-to-lymphocyte ratio, or potential platelet contamination, which may have influenced the observed expression patterns. The monocentric design may limit the generalizability of the findings. The deliberate 1:1 balancing of patients with and without thrombosis within each MPN subtype resulted in an enriched cohort that does not reflect real-world event rates. The study was not powered to detect differences between arterial and venous thrombotic events, and these subgroup analyses should therefore be interpreted with caution. Small subgroup sizes further limit statistical power, particularly for comparisons between usual and unusual site VTE. Future studies should focus on specific cell populations using comprehensive RNA sequencing and functional validation of the observed RNA expression changes. Moreover, further studies are warranted to clarify the role of selected HIF-1 α -regulated and thrombo-inflammatory genes and therapeutic potential.

Author contributions Study conception and design: OM, JR, AH, TE; collection of clinical data: OM, CC, AH; molecular analyses: TW, JR, TE; data analyses: OM, TW, JR; initial draft of the manuscript: OM; correction and approval of the final manuscript: all authors.

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Data Availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Competing Interests

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Therese Wolfrum has no conflict of interest.

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Thomas Ernst has no conflict of interest.

Jenny Rinke has no conflict of interest.

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Figure 1. Expression levels of target genes in leukocytes of patients with PV, ET and PMF.

Expression levels were compared between patients with and without thrombosis (A) in patients with polycythemia vera, (B) in patients with essential thrombocythemia, (C) in patients with primary myelofibrosis. *RPL13A* was used as a housekeeping gene for normalization of the data. Data analysis was performed according to the Pfaffl method [46]. *SLC2A1* - Solute Carrier Family 2 Member 1, *F3* - Tissue Factor, *SELP* - Selectin P, *VEGFA* - Vascular Endothelial Growth Factor A, *SERPINE1* - serpin peptidase inhibitor clade E member 1, *PROS1* - protein S, *NLRP3* - NLR family pyrin domain-containing 3, *THBS1* - thrombospondin 1. * $p < 0.05$, ** $p < 0.01$, ns (non-significant) $p \geq 0.05$.

Figure 2. RNA expression of target genes according to *JAK2* mutation status. 89 patients with V617F and one patient with exon 12 mutation. *RPL13A* was used as a housekeeping gene for normalization of the data. Data analysis was performed according to the Pfaffl method [46]. *SLC2A1* - Solute Carrier Family 2 Member 1, *F3* - Tissue Factor, *SELP* - Selectin P, *VEGFA* - Vascular Endothelial Growth Factor A, *SERPINE1* - serpin peptidase inhibitor clade E member 1, *PROS1* - protein S, *NLRP3* - NLR family pyrin domain-containing 3, *THBS1* - thrombospondin 1. * $p < 0.05$, ** $p < 0.01$, ns (non-significant) $p \geq 0.05$.

Figure 3. Correlation between *JAK2*-V617F VAF and expression levels of (A) *SLC2A1*, (B) *F3*, (C) *SELP*, (D) *VEGFA*, (E) *SERPINE1*, (F) *PROS1*, (G) *NLRP3*, and (H) *THBS1*. 89 patients with a *JAK2*-V617F mutation and a VAF $\geq 5.0\%$ were included. The ratio was calculated using the C_t value of the target gene and the C_t value of the housekeeping gene *RPL13A*. *SLC2A1* - Solute Carrier Family 2 Member 1, *F3* - Tissue Factor, *SELP* - Selectin P, *VEGFA* - Vascular Endothelial Growth Factor A, *SERPINE1* - serpin peptidase inhibitor clade E member 1, *PROS1* - protein S, *NLRP3* - NLR family pyrin domain-containing 3, *THBS1* - thrombospondin 1.

